Graphs and optimisation

Series 1

PRISTINI SCHOOL OF AI

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Exercise 1

1. Draw the diagram of each of the following graphs:

   a) $V = \{1, 2, 3, 4, 5, 6\}$ and $E = \{\{1, 2\}; \{1, 3\}; \{1, 4\}; \{2, 5\}; \{2, 6\}; \{3, 5\}; \{3, 6\}; \{4, 5\}; \{4, 6\}\}$

   b) $V = \{1, 2, 3, 4, 5\}$ and $E = \{\{1, 2\}; \{2, 3\}; \{2, 4\}; \{2, 5\}; \{3, 4\}; \{3, 5\}\}$

2. The **complement** of a graph $G = (V, E)$ is a graph $\overline{G} = (V, F)$ in which there is an edge between two vertices v and w if and only if there is no edge between v and w in G. Obviously, the complement of the complement of $\overline{G}$ is G.
   Draw the diagrams of the complements of the simple graphs identified in question 1.

Exercise 2

Consider the directed graph $G = (V, E, \Psi)$ such that:

$V = \{1, 2, 3, 4, 5\}$

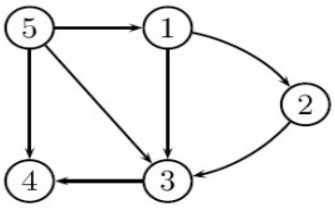
$E = \{(1, 2); (1, 4); (2, 2); (2, 3); (2, 4); (3, 5); (4, 3); (5, 3)\}$

1. Draw the diagram of graph G.
2. Represent in a table the incidence function Ψ.
3. Give the Out-degree of 2 and the In-degree of 4.
4. Give the set of predecessors of 4 and the set of successors of 2.
5. Give a partial graph and a sub-graph of G.

Exercise 3

Consider the following directed graph $G = (V, E, \Psi)$:

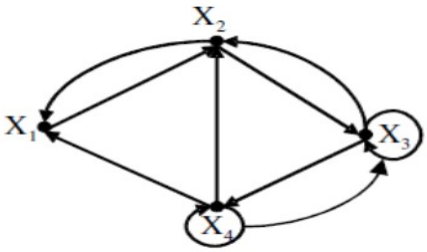
1. Define the sets V (set of vertices) and E (set of edges).
2. Give the order and the size of graph G.
3. Represent in a table the incidence function Ψ.



4. Is graph G planar? Justify your answer.
5. Which vertices are adjacent to vertex 1?
6. Determine the In-degrees and Out-degrees for each vertex.
7. Determine the vertices which are a source, a sink or an isolated point?

Exercise 4

Consider the following directed graph $G = (V, E, \Psi)$:



1. Define the sets V (set of vertices) and E (set of edges).
2. Specify the order and the size of graph G.
3. Represent in a table the incidence function Ψ.
4. Is graph G planar? Justify your answer.
5. Which vertices are adjacent to vertex 1?
6. Determine the In-degrees and Out-degrees for each vertex.
7. Determine the vertices which are a source, a sink or an isolated point?

Exercise 5

Three countries each send two spies to a conference; every spy mission is to spy on all the spies of other countries.

1. Represent this situation by a graph of order 6 in which each edge connecting i and j means that i spies on j, and j spies on i.

2. Is this graph complete ?

3. Is this graph connected ?

4. **a)** What is the degree of each vertex ?

   b) Deduce the number of edges of the graph.